



OPEN Prediction of potential high-temperature superconductivity in ternary Y–Hf–H compounds under high pressure

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Compressed ternary alloy superhydrides are currently considered to be the most promising competitors for high-temperature superconducting materials. Here, the stable stoichiometries in the Y–Hf–H ternary system under pressure are comprehensively explored in theory and four fresh phases are predicted: *Pmna*-YHfH₆ and *P4/mmm*-YHfH₇ at 200 GPa, *P4/mmm*-YHfH₈ at 300 GPa and *P-6m2*-YHfH₁₈ at 400 GPa. The four Y–Hf–H ternary phases are thermodynamically and dynamically stable at corresponding pressure. In addition, structural features, bonding characteristics, electronic properties, and superconductivity of the four ternary Y–Hf–H phases are systematically calculated and discussed. As the hydrogen content and the density of states of H atoms at the Fermi level increase, the superconducting transition temperatures (T_c) of Y–Hf–H system are significantly enhanced. The *P-6m2*-YHfH₁₈ with high hydrogen content exhibits a high calculated T_c value of 130 K at 400 GPa.

In recent years, the searches for superconducting materials have been a hotspot in the field of condensed matter physics, and realizing room-temperature superconductivity at atmospheric pressure is a long-held dream pursued by researchers. Hydrogen ranks first in the periodic table and has the smallest atomic mass in nature, thus exhibiting a substantially high Debye temperature¹. According to the Bardeen–Cooper–Schrieffer (BCS) theory, metallic hydrogen has been a prospective candidate for a room-temperature superconductor¹, which inspired extensive experimental and theoretical studies on the metallization of hydrogen under high pressure^{2–5}. However, the metallization of hydrogen requires extreme pressure over 500 GPa experimentally. In 2004, Ashcroft innovatively predicted that the lattice of hydrogen can be chemically “precompressed” to achieve metallization at lower pressures by alloying hydrogen with other elements⁶. This idea greatly reduces the pressure conditions required by room-temperature superconductors, which set off an upsurge in the investigation for room-temperature superconductivity of hydrogen-rich materials.

So far, the theoretical and experimental explorations for binary hydrides of almost all elements in the periodic table (except for some radioactive and magnetic elements) have been basically completed and considerable results have been achieved^{7–14}. Especially, the Y–H and Hf–H binary systems stand out due to their unusual hydrogen configuration structures and significantly high superconducting transition temperatures under pressures. For hafnium hydrogen compounds, the dense superhydrides HfH₁₀ with a planar “pentagraphenelike” sublattice can exhibit a predicted T_c up to 234 K at 250 GPa¹⁵. HfH₉ with H₁₂ tube and HfH₃ framework has a predicted T_c of 110 K at 200 GPa¹⁶. A superconducting critical temperature with $T_c \sim 83$ K was observed in HfH₁₄ at 243 GPa in the experiment¹⁷. For binary yttrium hydrides, the calculated T_c of YH₆ with H₂₄ cage-like clathrate structure achieves 251–264 K at 120 GPa¹⁸ and 290 K at 300 GPa¹⁹. Based on the calculation, YH₉ with H₂₉-cage structure has a T_c of 276 K at 150 GPa²⁰. YH₁₀ with H₃₂-cage structure possesses an T_c value of 303 K at 400 GPa²⁰ in theoretical aspect. Among them, the high-temperature superconductivity of YH₆ and YH₉ materials have been confirmed in subsequent high-pressure experiments^{21–23}.

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The general rules of hydrogen-rich superconductors are summarized based on binary hydrides and applied to ternary hydrides with more abundant hydrogen configurations by the diverse chemical compositions^{24–29}. The explorations on ternary hydrogen-rich materials can be mainly divided into two aspects. (i) Add another element to the binary hydride for structural modulation to form a new ternary hydride with better superconductivity. For example, the designed $\text{Li}_2\text{MgH}_{16}$ with remarkably high predicted T_c of 473 K at 250 GPa via doping lithium into the parent MgH_{16} ³⁰. (ii) Mix metal binary hydrides proportionally in the form of solid solutions to obtain alloy-type ternary hydrides, in which the metal elements have similar electronegativity and atomic radius. Theoretical predictions show that ternary hydrides such as $(\text{Y}, \text{Sr})\text{H}_{11}$ ³¹, YCaH_{12} ³², and YZrH_{18} ³³ through alloying hydrogen-rich binary compounds are estimated to exhibit good superconductivity under high pressures, and the experimentally observed T_c values of $(\text{La}, \text{Y})\text{H}_{10}$ ³⁴, $(\text{La}, \text{Y})\text{H}_6$, and $(\text{Y}, \text{Ce})\text{H}_9$ ³⁵ at various pressures are 253 K, 226 K, and 140 K, respectively. Obviously, ternary hydrides have made significant breakthroughs in the discovery of room-temperature superconductors both theoretically and experimentally.

It is widely known that Y and Hf elements are located diagonally in the periodic table with close electronegativity, which indicates prominent high-temperature superconductivity potentially exists in Y–Hf–H system. In our work, we perform an extensive calculation to investigate the ternary Y–Hf–H system under high pressure and analyze their thermodynamical stability and dynamical stability. Indeed, several new stoichiometries in the ternary Y–Hf–H system have been predicted under pressure. The structural features, Bader charges, band structures, density of states (DOS), electron–phonon coupling (EPC) and superconducting transition temperatures of the new ternary Y–Hf–H phases are systematically studied and they are found to possess the unusual structural features and superconductivity.

Results and discussion

Ternary phase diagrams and phase stability

In the first step, in order to determine the stable low-energy compounds of Y–Hf–H ternary system under high pressure, we thoroughly explored the potential stoichiometries of $\text{Y}_x\text{Hf}_y\text{H}_z$ ($x = 1–3$, $y = 1–3$, $z = 2–24$) at pressures of 200, 300 and 400 GPa, respectively, and constructed the ternary phase diagrams of Y–Hf–H system under selected pressures (see Fig. 1). Its corresponding stable elemental solids (Y, Hf and H) located at three vertexes of triangle and binary hydrides (Y–H and Hf–H systems) on three sides have been systematically investigated experimentally and theoretically. The elemental solids and binary compounds information referenced in the present structural predictions are from previous works^{15–23,36–39}. From Fig. 1a, YHfH_6 and YHfH_7 are both thermodynamic stable stoichiometries, which are exactly located on the convex hull of 200 GPa. Except for directly compressing Y, Hf, and H, the two red lines exhibit others potential synthetic routes for YHfH_6 ($\text{YH}_3 + \text{HfH}_3 \rightarrow \text{YHfH}_6$) and YHfH_7 ($\text{YH}_4 + \text{HfH}_3 \rightarrow \text{YHfH}_7$) based on the “triangle straight-line method” (TSLM)^{40,41}. As the pressure increased, YHfH_6 gradually becomes thermodynamic unstable while YHfH_7 can maintain energy stability up to 300 GPa, and a new stable component YHfH_8 has emerged on the convex hull. Similarly, YHfH_8 can be synthesized through the route $\text{YH}_4 + \text{HfH}_4 \rightarrow \text{YHfH}_8$ shown in the red line of Fig. 1b. Upon further compression, YHfH_{18} compound with the highest hydrogen content, as the only composition falling into the convex hull, can be stable energetically superior to other stoichiometries at 400 GPa (see Fig. 1c). The above analysis demonstrates that YHfH_6 , YHfH_7 , YHfH_8 and YHfH_{18} are thermodynamically stable under high pressures. We also provide their potential synthetic routes, which will stimulate the experimental synthesis. Subsequently, we focus on the structural characteristics, electronic properties, and superconductivity of YHfH_6 and YHfH_7 at 200 GPa, YHfH_8 at 300 GPa, and YHfH_{18} at 400 GPa, respectively.

Structural features, bonding features, and electronic properties

The crystal structure diagrams of YHfH_6 and YHfH_7 at 200 GPa, YHfH_8 at 300 GPa and YHfH_{18} at 400 GPa are depicted in Fig. 2 and homologous detailed structural parameters are summarized in Table S1 of Supplementary Information (SI). In addition, to better understand the interaction information, we also depict their electronic localization functions (ELF) in Fig. 3. The predicted YHfH_6 is found to crystallize in the orthorhombic $Pmna$ phase with the lattice parameters of $a = 10.170$ Å, $b = 8.108$ Å, and $c = 8.209$ Å at 200 GPa. This phase is observed in ternary hydrides for the first time. There are three inequivalent H atoms are located at the Wyckoff sites $4h$ (0.000, 0.360, 1.386), $4h$ (0.000, 0.124, 1.123) and $4g$ (0.250, 0.239, 0.750), respectively, and these H atoms form

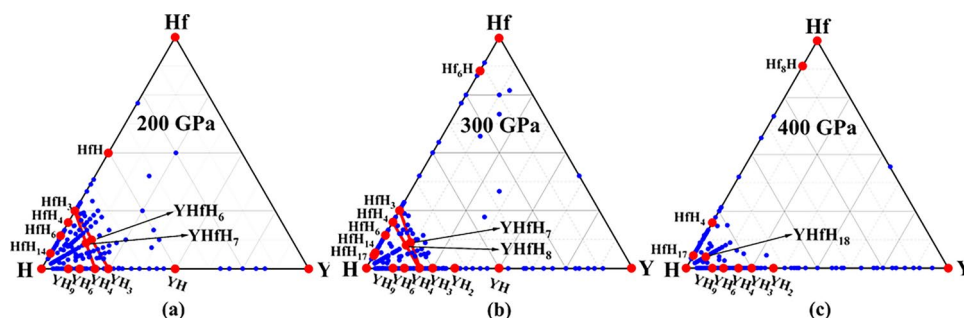


Figure 1. Convex hulls of Y–Hf–H system at (a) 200 GPa, (b) 300 GPa, and (c) 400 GPa. Red and blue dots represent thermodynamically stable and unstable structures, respectively.

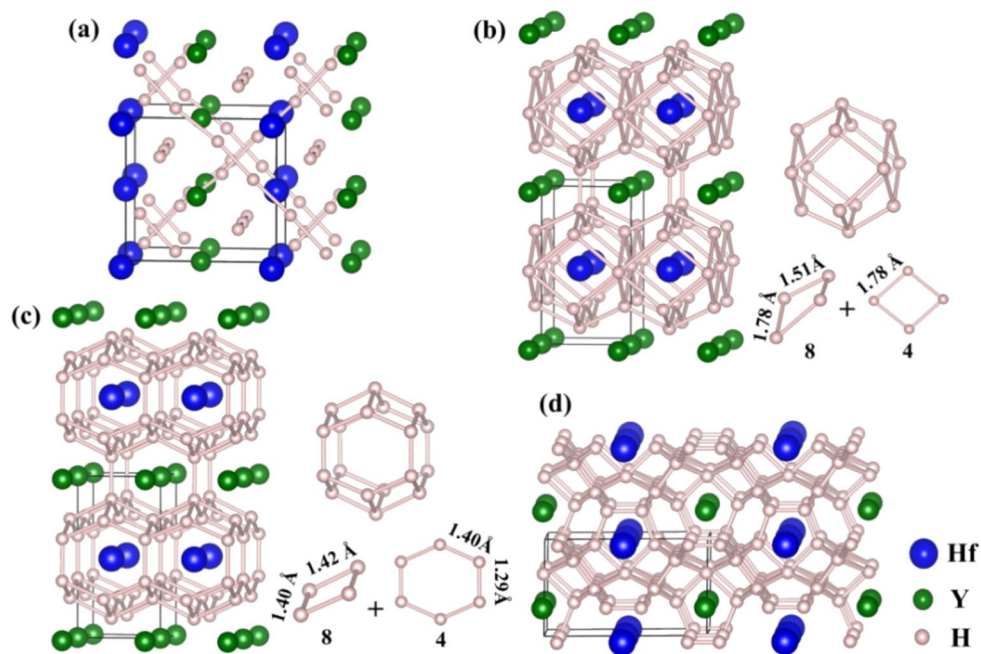


Figure 2. Crystal structures of (a) $Pmna$ -YHfH₆ at 200 GPa, (b) $P4/mmm$ -YHfH₇ at 200 GPa, (c) $P4/mmm$ -YHfH₈ at 300 GPa and (d) $P-6m2$ -YHfH₁₈ at 400 GPa. The blue, green and pink spheres represent Hf, Y and H atoms, respectively.

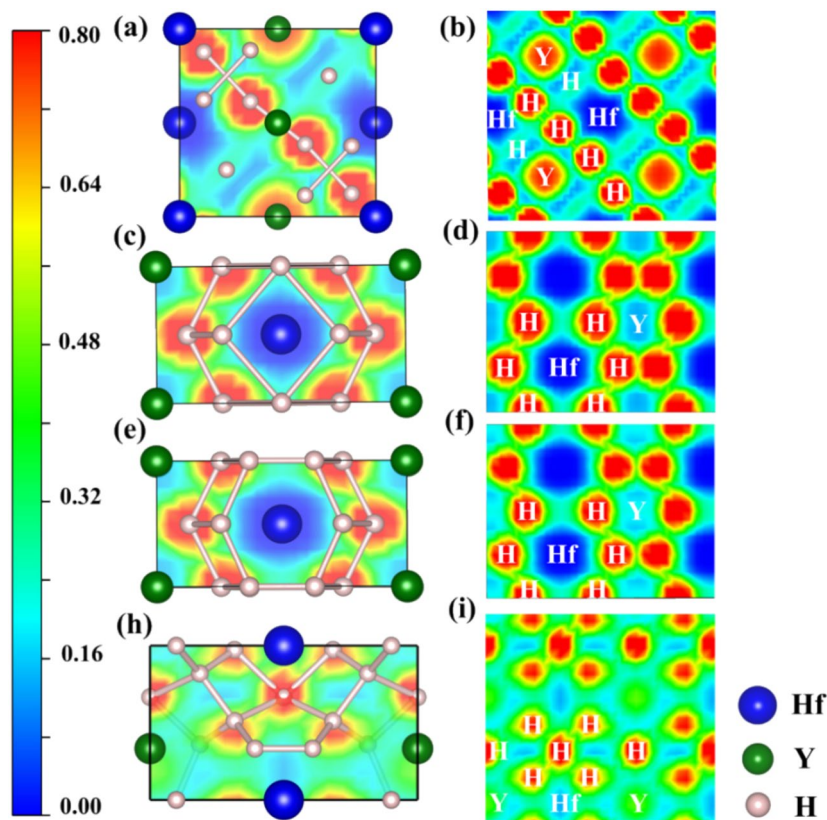


Figure 3. Electronic localization functions (ELF) of (a,b) $Pmna$ -YHfH₆ at 200 GPa, (c,d) $P4/mmm$ -YHfH₇ at 200 GPa, (e,f) $P4/mmm$ -YHfH₈ at 300 GPa and (h,i) $P-6m2$ -YHfH₁₈ at 400 GPa. The blue, green and pink spheres represent Hf, Y and H atoms, respectively.

one-dimensional linear H₄ chains with two types H–H separations of 1.626 and 1.669 Å, as shown in Fig. 2a. Its weak covalence is accurately described by the ELF in Fig. 3a,b, while the absence of electron localization between metal atoms and H atom reveals the purely ionic of neighboring Hf–H and Y–H connections. For YHfH₇ at 200 GPa and YHfH₈ at 300 GPa, both are stable with tetragonal *P4/mmm* phase, which adopts the same symmetry as previously reported YZrH₈³³, YPrH₈⁴², YBaH₈⁴³ and YMgH₈⁴⁴ hydrides. In this structure, Hf atoms occupy the center position of the tetragonal lattice while Y atoms are located at eight vertices, and all H atoms form H-clathrate structures around metal Hf atoms. The Hf-centered H₁₄ cage in *P4/mmm*-YHfH₇ contains eight twisted quadrilateral and four square “H₄” units with only two types H–H distances of 1.51 and 1.78 Å (see Fig. 2b). The hydrogen sublattice of *P4/mmm*-YHfH₈ is very similar to YHfH₇, which can be considered to insert an H atom into each long edges of YHfH₇ lattice, forming an H₁₈ cage with three types H–H separations of 1.29, 1.40 and 1.42 Å (see Fig. 2c). It is noteworthy that this H atoms insertion facilitates the four squares “H₄” of *P4/mmm*-YHfH₇ to evolve into four hexagon “H₆” units in *P4/mmm*-YHfH₈. The shortest H–H distance 1.29 Å in “H₆” units are greatly shortened compared to that in “H₄” (1.78 Å), which makes it closer to the H–H distance (0.98 Å) in metallic monatomic H at 500 GPa⁴⁵. The decrease in distance between H atoms enhances their mutual interactions, which not only promotes the stability of the hydrogen cage skeleton, but also plays a decisive role in the superconductivity of the system. From Fig. 3c–f, the H atoms are weakly covalently bonded to one another in the cages, which are evident in view of the charge localizations between the nearest-neighbor H atoms. In the case of YHfH₁₈ at 400 GPa, it is stable in hexagonal structure with *P-6m2* symmetry, which has the lattice parameters of *a* = *b* = 3.176 Å and *c* = 4.770 Å (see Fig. 2d). It is found that all H atoms form semi-cages with the H–H connections around 1.03–1.46 Å in this phase. The weak covalence between H–H connection is also demonstrated by the ELF diagrams plotted in Fig. 3h,i, as the ELF values between metal atoms and the nearest H atom are smaller than 0.5. In addition, the shortest H–H bond length in *P-6m2*-YHfH₁₈ phase is closest to the atomic H under 500 GPa⁴⁵ and famous ultra-high T_c hydride Li₂MgH₁₆ (1.02 Å, at 300 GPa)³⁰, HfH₁₀ (1.14 Å, at 200 GPa)¹⁵, YH₁₀ (1.08 Å, at 200 GPa)²⁰ and LaH₁₀ (1.11 Å, at 200 GPa)⁴⁶, etc.

The Bader charge analysis⁴⁷ was conducted to further gain insight into the chemical bonding features between Y, Hf and H atoms in *Pmna*-YHfH₆, *P4/mmm*-YHfH₇, *P4/mmm*-YHfH₈ and *P-6m2*-YHfH₁₈, as tabulated in Table 1. As the electronegativity of Y (1.2) and Hf (1.3) are much smaller than H (2.2), a large amount of charge should transfer from Y and Hf atoms to H atoms based on the periodic law⁴⁸. The calculated results show that each Hf atom approximately loses 1.21, 1.25, 1.26 and 1.26 *e* electrons in YHfH₆, YHfH₇, YHfH₈ and YHfH₁₈, respectively, and one Y atom loses 1.23, 1.32, 1.27 and 1.32 *e* charges, respectively. For H atoms, per H gains charges of 0.41, 0.37, 0.32 and 0.14 *e*, respectively. We intuitively observe that the number of electrons accepted by H atoms gradually decreases in these four structures with the rise of pressure. The obtained electrons for H–H unit will reside in the H–H antibonding orbitals, which can elongate the H–H bond length in the high-pressure hydrides^{49,50}. Consequently, the H–H bonds in four phases present a gradually decreasing trend as well. The discussion of Bader charge is consistent with the above ELF analysis, further confirming the strong ionic natures between metal atoms and H atoms in the four stable configurations.

The electronic band structures and DOS of *Pmna*-YHfH₆, *P4/mmm*-YHfH₇, *P4/mmm*-YHfH₈ and *P-6m2*-YHfH₁₈ at high pressure were calculated to reveal their electronic properties, as exhibited in Fig. 4. As a result, these four structures possess metallic properties due to the overlapping of the conduction and valence bands, and a large total density of states (TDOS) at the Fermi level also confirms this conclusion. Furthermore, it is found that the electronic density of states at Fermi level is obviously dominated by metal Y and Hf atoms in YHfH_{*n*} (*n* = 6, 7 and 8) phases, so the metallic properties of the above three phases are contributed by metal atoms. Due to their less hydrogen content, the H atoms provide low total H-DOS with 0.21, 0.11 and 0.16 (states/eV)/f.u., respectively, and H-DOS merely account for 11%, 15% and 23% of the total density of states, respectively. Interestingly, the electronic band structures of YHfH₇ and YHfH₈ are quite similar, which is attributed to their similar crystal structures. As for *P-6m2*-YHfH₁₈ phase with the highest H content at 400 GPa, more electrons participate in the formation of Cooper pairs and there is a steep large electron pocket passing through

Pressure (GPa)	Space group	Atom	Charge value (e)	δ (e)
200	<i>Pmna</i> -YHfH ₆	Hf	2.79	1.21
		Y	9.77	1.23
		H	1.41	−0.41
200	<i>P4/mmm</i> -YHfH ₇	Hf	2.75	1.25
		Y	9.68	1.32
		H	1.37	−0.37
300	<i>P4/mmm</i> -YHfH ₈	Hf	2.74	1.26
		Y	9.73	1.27
		H	1.32	−0.32
400	<i>P-6m2</i> -YHfH ₁₈	Hf	2.74	1.26
		Y	9.68	1.32
		H	1.14	−0.14

Table 1. Calculated Bader charges of Hf, Y and H atoms in four stable phases. δ represents the amount of charge transferred from Hf and Y atoms to H atom.

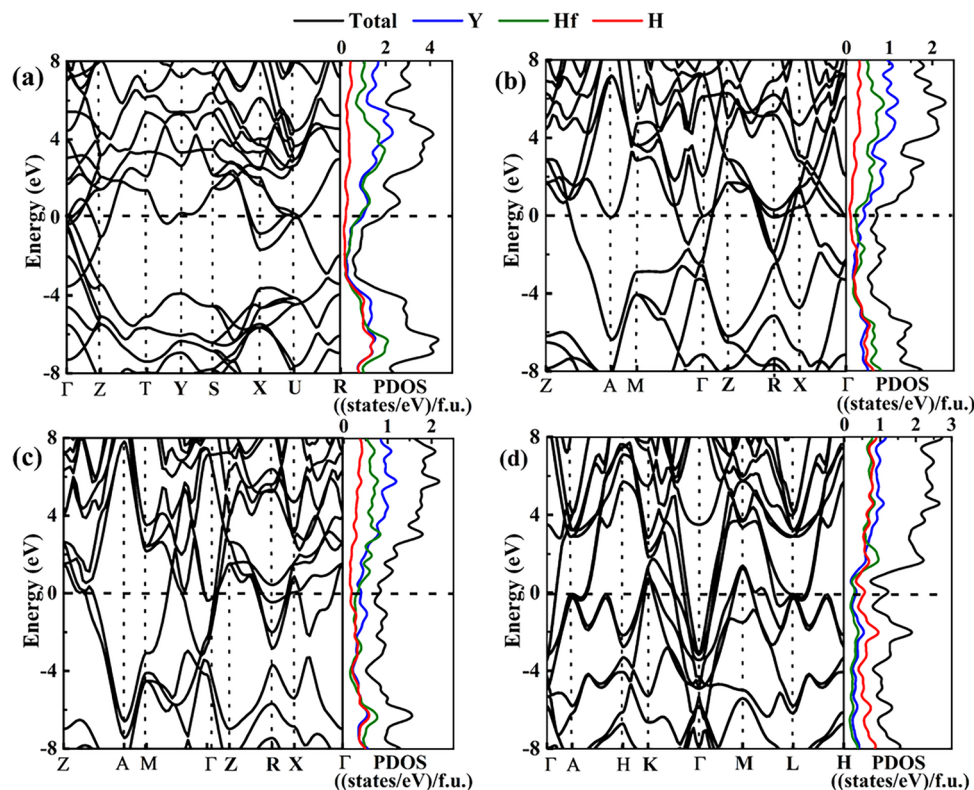


Figure 4. Electronic band structures and density of states (DOS) of (a) *Pmna*-YHfH₆ at 200 GPa, (b) *P4/mmm*-YHfH₇ at 200 GPa, (c) *P4/mmm*-YHfH₈ at 300 GPa and (d) *P-6m2*-YHfH₁₈ at 400 GPa.

the Fermi level at Γ point, which increases the TDOS at Fermi level. In this structure, the total H-DOS at Fermi level significantly enhances to 0.67 (states/eV)/f.u., which is comparable to HfH₉¹⁶, YH₆¹⁸, YCa₂H₁₈⁵¹, YCaH₂₀⁵¹, CaHfH₁₈⁵², YLuH₁₂⁵³ and Rb₂MgH₁₈⁵⁴, and the proportion of H-DOS in TDOS increases to 47% as well. In fact, the H-derived DOS at Fermi level is crucial for considerably improving the superconductivity of *P-6m2*-YHfH₁₈ in Y–Hf–H ternary system and this will be demonstrated in the subsequent discussions.

Dynamical properties and electron–phonon coupling

The computed phonon dispersion curves, projected phonon density of states (PHDOS), Eliashberg spectral function $\alpha^2F(\omega)$ and the integral EPC parameters $\lambda(\omega)$ of YHfH₆, YHfH₇, YHfH₈ and YHfH₁₈ are plotted in Fig. 5. The phonon dispersion curves can effectively confirm their dynamic stability because of the absence of any imaginary phonon frequencies throughout the whole Brillouin zone. In the PHDOS, the low-frequency phonon bands are mainly originated from the vibrations of Hf and Y atoms, while the middle-frequency and high-frequency phonon modes regions are associated with H atoms due to the Hf and Y atomic masses are much heavier than H atoms.

Based on the discussion of electronic properties of four stable phases, their excellent metallic properties under pressures prompt us to explore their potential superconductivity. Therefore, the electron–phonon coupling parameters λ , the logarithmic average phonon frequency $\omega_{\log}$ and electron density of states at Fermi level $N(E_F)$ were calculated in Table 2. Moreover, based on the Eliashberg spectral function $\alpha^2F(\omega)$ and the integral EPC parameters $\lambda(\omega)$ of *Pmna*-YHfH₆ at 200 GPa, *P4/mmm*-YHfH₇ at 200 GPa, *P4/mmm*-YHfH₈ at 300 GPa and *P-6m2*-YHfH₁₈ at 400 GPa in Fig. 5, the superconducting critical temperature values are evaluated through solving the Allen–Dynes modified McMillan equation⁵⁵:

$$T_c = \frac{\omega_{\log}}{1.2} \exp \left[\frac{1.04(1 + \lambda)}{\lambda - \mu^*(1 + 0.62\lambda)} \right].$$

For the hydrides with $\lambda < 1.5$, this equation can estimate highly accurate T_c values. Among them, the Coulomb pseudopotential constants μ^* are set to typical 0.10 and 0.13, respectively. It is found that the T_c values slightly decrease when $\mu^* = 0.13$.

In YHfH₆ and YHfH₇ phases at 200 GPa, their calculated EPC integral coefficients λ are close (~ 0.38), which results in their similar predicted T_c values around 3 K (see Table 2). Although YHfH₆ has a large electronic density of states at the Fermi level [$N(E_F) = 11.983$ states/spin/Ry/Unit Cell], even the largest among the four stable structures, this large $N(E_F)$ value is mainly provided by metal Hf and Y atoms, which suppresses the superconductivity of YHfH₆. From Fig. 5, the low-frequency heavy Hf and Y atomic vibrations below 6 THz contribute 51% of the total λ , while the high-frequency region above 20 THz associated with light H atom vibration accounts

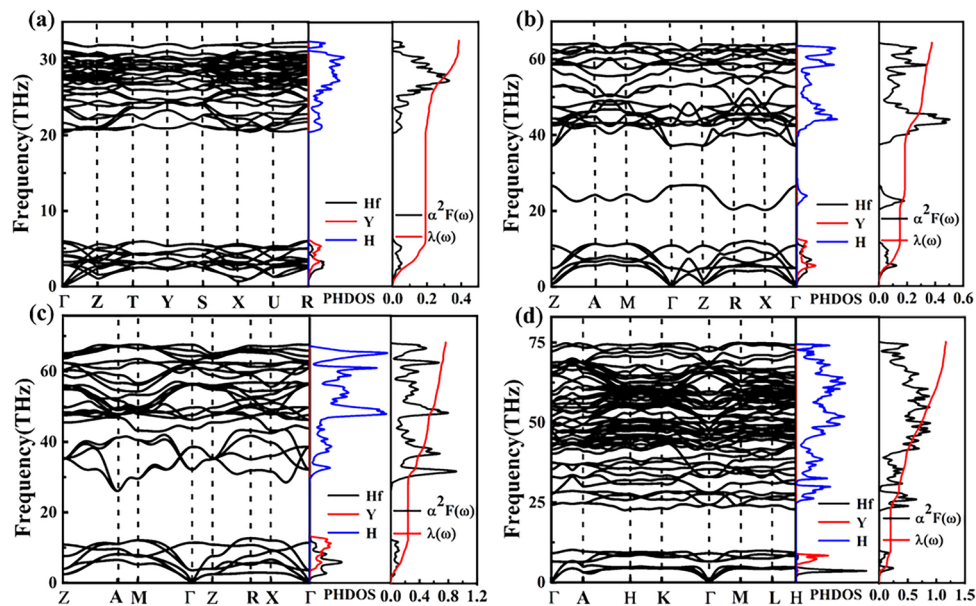


Figure 5. The phonon dispersion curves, the projected phonon density of states and Eliashberg spectral function $\alpha^2F(\omega)$ together with the electron–phonon integral $\lambda(\omega)$ of (a) *Pmna*-YHfH₆ at 200 GPa, (b) *P4/mmm*-YHfH₇ at 200 GPa, (c) *P4/mmm*-YHfH₈ at 300 GPa and (d) *P-6m2*-YHfH₁₈ at 400 GPa.

Structure	Pressure (GPa)	$\omega_{\log}$	$N(E_f)$	λ	T_c	
					$\mu^*=0.10$	$\mu^*=0.13$
<i>Pmna</i> -YHfH ₆	200	849.375	11.983	0.380	2.613	1.013
<i>P4/mmm</i> -YHfH ₇	200	944.812	4.189	0.381	2.995	1.484
<i>P4/mmm</i> -YHfH ₈	300	1116.317	5.233	0.760	46.668	36.976
<i>P-6m2</i> -YHfH ₁₈	400	1463.220	6.538	1.196	130.465	115.663

Table 2. Calculated electron–phonon coupling parameters λ , electronic density of states at the Fermi level $N(E_f)$ (states/spin/Ry/Unit Cell), logarithmic average phonon frequency $\omega_{\log}$ (K), and superconducting transition temperatures T_c (K) of *Pmna*-YHfH₆ at 200 GPa, *P4/mmm*-YHfH₇ at 200 GPa, *P4/mmm*-YHfH₈ at 300 GPa and *P-6m2*-YHfH₁₈ at 400 GPa.

for 49% in YHfH₆. As for YHfH₇ structure, its phonon modes can be divided into three main regions. The low frequency phonon modes below 10 THz provided by metal atoms account for 39%, while H atoms drive 10% and 51% in the middle (23–27 THz) and upper (> 40 THz) parts, respectively. In this situation, the contribution of the H-phonon band begins to increase. For YHfH₈ at 300 GPa, the dominance of H atom in electron–phonon coupling is further clarified and the contribution of the H-phonon band enhances to 68%. In addition, the significant difference in superconductivity between *P4/mmm*-YHfH₇ and *P4/mmm*-YHfH₈ with highly similar structures can be further attributed to the fact that the H₁₈ cage of YHfH₈ bonds more tightly than the H₁₄ cage in YHfH₇. Based on these reasons, the T_c

value of *P4/mmm*-YHfH₈ has been effectively enhanced to 47 K. Notably, in YHfH₁₈ phases at 400 GPa, metal atoms motions in the lowest part of the phonon bands (< 10 THz) only occupy 18% of the total λ , while all other vibrational modes up to 82% (> 25 THz) are dominated by H atoms. Obviously, YHfH₁₈ possess the highest predicted T_c value 130 K among the four stable structures, which is close to the theoretical predictions of other H-rich materials such as YZrH₁₈ (156 K)³³ and LiP₂H₁₄ (169 K)⁵⁶, but higher than the predicted results of MgH₁₂ (60 K)⁵⁷, YCH₁₂ (112 K)⁵⁸, YSH₆ (91 K)⁵⁹, Ca₂B₂H₁₃ (89 K)⁶⁰ and Mg(ScH₄)₃ (10 K)⁶¹. This result highlights that the high-frequency H atom vibration exhibits a significant contribution to the electron–phonon coupling in YHfH₁₈ structure, which is coincide with the analysis result of large H-DOS proportion at the Fermi level. Thus, we have certificated that H-derived DOS at the Fermi level can effectively improve the T_c value of Y–Hf–H ternary hydrides under high pressure.

Conclusions

In conclusion, we conducted comprehensive structural searches of the Y–Hf–H ternary system under high pressure by applying unbiased CALYPSO method and first-principles calculations to reveal the phase stability. As a result, we have predicted four thermodynamically stable ternary hydrogen-rich compounds at selected pressures, namely *Pmna*-YHfH₆, *P4/mmm*-YHfH₇, *P4/mmm*-YHfH₈ and *P-6m2*-YHfH₁₈. YHfH₆ contains linear H₄ chains, while YHfH₇ and YHfH₈ adopt clathrate structures of H₁₄ and H₁₈ cages, respectively. Additionally,

YHfH₁₈ forms intriguing H semi-cages lattices. In these four stable structures, ionic bonds are formed between the metal and H atoms, and the two nearest H atoms are connected by weak covalent bonds. According to their metallic properties, the calculated superconducting transition temperatures of YHfH₈ and YHfH₁₈ are 47 K at 300 GPa and 130 K at 400 GPa, respectively. We believe that the density of states at the Fermi level dominated by H atoms are closely associated with the high-temperature superconductivity of the Y–Hf–H system.

Computational methods

The extensive variable-composition structure searches of Y–Hf–H ternary system under high pressures were implemented via swarm intelligence-based CALYPSO (Crystal structure AnaLYsis by Particle Swarm Optimization) code^{62,63} including 1–2 formula units (f.u.) at 200, 300 and 400 GPa. A similar method has successfully predicted the high-pressure structures of various systems^{64–69} and provides effective guidance for experimental synthesis. We have performed the structural relaxations, energetic and electronic structures in the VASP (Vienna ab initio simulation package) code⁷⁰ through density functional theory (DFT) within the Perdew–Burke–Ernzerhof (PBE) exchange correlation function of generalized gradient approximation (GGA)^{71,72}. The Projector-Augmented Wave (PAW)⁷³ method with valence electron configurations 4s²4p⁶4d¹5s² for Y, 6s²5d² for Hf and 1s¹ for H was adopted with 800 eV cutoff energy for the plane-wave basis expansion of the electronic wave functions and appropriate Monkhorst–Pack scheme with a *k*-point grid of $2\pi \times 0.03 \text{ \AA}^{-1}$ for sampling in the Brillouin zone to ensure that all the enthalpy calculations were well converged to less than 1 meV/atom⁷⁴. The descriptions of electronic properties and bonding characteristics have been conducted through Bader charge analysis⁴⁷ and ELF method^{75,76}. Additionally, the phonon and electron–phonon coupling matrix elements were calculated within density functional perturbation theory (DFPT) as implemented in the QUANTUM ESPRESSO package⁷⁷ with a kinetic cutoff energy of 80 Ry. In the first Brillouin zone, we have selected *k*-point meshes of $12 \times 12 \times 12$, $20 \times 20 \times 12$, $16 \times 16 \times 16$ and $20 \times 20 \times 16$ for *Pmna*-YHfH₆, *P4/mmm*-YHfH₇, *P4/mmm*-YHfH₈ and *P-6m2*-YHfH₁₈, respectively, and *q*-point meshes of $3 \times 3 \times 3$, $5 \times 5 \times 3$, $4 \times 4 \times 4$ and $5 \times 5 \times 4$ for *Pmna*-YHfH₆, *P4/mmm*-YHfH₇, *P4/mmm*-YHfH₈ and *P-6m2*-YHfH₁₈, respectively.

Data availability

The datasets used and/or analyzed during the current study available from the corresponding author on reasonable request.

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Author contributions

Y.Q.W., Y.Y.J. and C.Z.Z. conceived the idea. Y.Q.W., F.L.Y., J.Q.Z., S.L. and S.B.C. performed the calculations. Y.Q.W., Y.Y.J. and C.Z.Z. wrote the manuscript and all authors contributed to revisions.

Competing interests

The authors declare no competing interests.

Additional information

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